

MATH 3210-004 PSet 6

Specification

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1 Background

This problem set focuses on integral calculus; specifically all but the last problem study **integration according to Riemann**. The formal definition of Riemann integrability is rather technical, and the problems in this problem set are mostly about the foundational properties of Riemann integrability and Riemann integrals. The last problem then studies a special class of functions, namely those that are of bounded variation.

Summary of Notation For your convenience here is a quick summary of the notation introduced either in previous problem sets or in class; the same notations are used in the problems below. For $I = [\lambda, \rho]$ a compact interval,

- $\text{Par}(I)$ is the collection of all (finite interval) partitions of I .
- For any $\mathcal{P} \in \text{Par}(I)$, $S(\mathcal{P})$ denotes the set of all sampling functions of $\mathcal{P}$; thus by definition $s \in S(\mathcal{P})$ iff $s : \mathcal{P} \rightarrow I$ is a function such that for any $P \in \mathcal{P} : s(P) \in P$.

- $s\text{Par}(I)$ is the collection of all sampled (finite interval) partitions of I ; thus any element in $s\text{Par}(I)$ is of the form $(\mathcal{P}, s)$, where $\mathcal{P} \in \text{Par}(I)$ and $s \in S(\mathcal{P})$
- For any $\mathcal{P} \in \text{Par}(I)$ and for any $P \in \mathcal{P}$, $\ell(P)$ denotes the length of the interval P , that is, $\ell(P) = \sup(P) - \inf(P)$.
- $F(I; \mathbb{R})$, $F_b(I; \mathbb{R})$, $R(I; \mathbb{R})$, $B(I; \mathbb{R})$, $C^0(I; \mathbb{R})$ denote respectively the sets of all, bounded, Riemann integrable, Bourbaki integrable (see the background section of [problem set 5](#)) and continuous real valued functions on I .
- For any $f \in F(I; \mathbb{R})$, and for any $(\mathcal{P}, s) \in s\text{Par}(I)$ the Riemann sum of f relative to $(\mathcal{P}, s)$ is defined by

$$R(f; \mathcal{P}, s) = \sum_{P \in \mathcal{P}} f(s(P))\ell(P) \in \mathbb{R}.$$

- For any $f \in F_b(I; \mathbb{R})$ and for any $\mathcal{P} \in \text{Par}(I)$, the lower Darboux sum, upper Darboux sum and Darboux aperture of f relative to $\mathcal{P}$ are defined respectively by

$$L(f; \mathcal{P}) = \sum_{P \in \mathcal{P}} \inf(f|_P) \ell(P) \in \mathbb{R},$$

$$U(f; \mathcal{P}) = \sum_{P \in \mathcal{P}} \sup(f|_P) \ell(P) \in \mathbb{R},$$

$$D(f; \mathcal{P}) = U(f; \mathcal{P}) - L(f; \mathcal{P}) = \sum_{P \in \mathcal{P}} \text{diam}(f(P))\ell(P) \in \mathbb{R}_{\geq 0}.$$

Compatibility Let I be a compact interval, $\sigma \in F(I; \mathbb{R})$ be a step function and $\mathcal{P} \in \text{Par}(I)$. σ and $\mathcal{P}$ are said to be **compatible** if

$$\forall P \in \mathcal{P}, \exists y \in \text{im}(\sigma), \forall x \in]\inf(P), \sup(P)[: \sigma(x) = y.$$

(In words, σ and $\mathcal{P}$ are compatible if on each $P \in \mathcal{P}$, σ is constant except possibly at the endpoints.)

Bounded Variation Let I be a compact interval, $f \in F(I; \mathbb{R})$. f is said to be of **bounded variation** if

$$\sup_{\mathcal{P} \in \text{Par}(I)} \sum_{P \in \mathcal{P}} |f(\sup(P)) - f(\inf(P))| < \infty.$$

We denote the collection of all real valued functions of bounded variation on I by $BV(I; \mathbb{R})$.

Refinement Let I be a compact interval, $\mathcal{P}, \mathcal{Q} \in \text{Par}(I)$. We say that $\mathcal{P}$ **refines** $\mathcal{Q}$ if for any $P \in \mathcal{P}$, there is a $Q \in \mathcal{Q}$ such that $P \subseteq Q$. If $\mathcal{P}$ refines $\mathcal{Q}$, we write

$$\boxed{\mathcal{P} \succ_{\text{ref}} \mathcal{Q}}.$$

For $(\mathcal{P}, s), (\mathcal{Q}, t) \in \text{sPar}(I)$, if $\mathcal{P} \succ_{\text{ref}} \mathcal{Q}$, we again say that $(\mathcal{P}, s)$ **refines** $(\mathcal{Q}, t)$ and write

$$\boxed{(\mathcal{P}, s) \succ_{\text{ref}} (\mathcal{Q}, t)}.$$

Mesh Let I be a closed and bounded interval, $\mathcal{P}, \mathcal{Q} \in \text{Par}(I)$. The **mesh** of $\mathcal{P}$ is the nonnegative number $\text{mesh}(\mathcal{P}) = \sup_{P \in \mathcal{P}} \ell(P)$. If $\text{mesh}(\mathcal{P}) \leq \text{mesh}(\mathcal{Q})$, we write

$$\boxed{\mathcal{P} \succ_{\text{mesh}} \mathcal{Q}}.$$

For $(\mathcal{P}, s), (\mathcal{Q}, t) \in \text{sPar}(I)$, if $\mathcal{P} \succ_{\text{mesh}} \mathcal{Q}$, we again write

$$(\mathcal{P}, s) \succ_{\text{mesh}} (\mathcal{Q}, t).$$

2 Generative AI and Computer Algebra Systems Regulations

This section applies only if you choose to use either a **generative AI tool** (e.g. chatbots) or a **computer algebra system (CAS)** while working on this problem set. If you do not use such tools, you may skip this section.

The use of these tools while completing this problem set is permitted, provided it is done responsibly and in a manner that supports your learning. Note however that in exams you will not be allowed to use any AI or CAS.

2.1 Disclosure

If you use such tools, you must disclose this fact in the designated section of the form you will complete as part of your submission. You must also share your reflections on which parts of the problem required your own mathematical judgment, insight, or decision-making, and which parts could reasonably be delegated to a computational or generative tool.

2.2 Guidelines for Responsible Use

If you use a generative AI tool or chatbot, you must adhere to the following guidelines:

- During the chat you may share with the chatbot parts or all of this specification document, as well as parts of the textbook or other sources.
- Directly asking the tool for complete problem solutions is prohibited.
- Appropriate uses include asking for conceptual explanations, checking intermediate steps, clarifying definitions or theorems, or exploring alternative solution approaches, provided that the final work submitted is your own.
- Regardless of the tools used, you are expected to understand, justify, and be able to reproduce all submitted work independently. The use of AI or CAS does not reduce or replace this expectation.

3 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details.

When tackling statements in MATH 3210, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.

3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example, you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. Such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let I be a compact interval. Then

- (a) For any step function $\sigma \in F(I; \mathbb{R})$, there is a $\mathcal{P} \in \text{Par}(I)$ such that σ and $\mathcal{P}$ are compatible.
- (b) For any $\mathcal{P} \in \text{Par}(I)$, there is a step function $\sigma \in F(I; \mathbb{R})$ such that σ and $\mathcal{P}$ are compatible.
- (c) For any $\mathcal{P} \in \text{Par}(I)$, there is a step function $\sigma \in F(I; \mathbb{R})$ such that σ and $\mathcal{P}$ are compatible and

$$L(f; \mathcal{P}) = \int_I \sigma(x) dx.$$

- (d) For any $\mathcal{P} \in \text{Par}(I)$, there is a step function $\tau \in F(I; \mathbb{R})$ such that τ and $\mathcal{P}$ are compatible and

$$U(f; \mathcal{P}) = \int_I \tau(x) dx.$$

- (e) For any $(\mathcal{P}, s) \in \text{sPar}(I)$, there is a step function $v \in F(I; \mathbb{R})$ such that v and $\mathcal{P}$ are compatible and

$$R(f; \mathcal{P}, s) = \int_I v(x) dx.$$

2. Let I be a compact interval. Then

- (a) $\succsim_{\text{ref}}$ is a **directed set structure** on both $\text{Par}(I)$ and $\text{sPar}(I)$.
- (b) $\succsim_{\text{mesh}}$ is a directed set structure on both $\text{Par}(I)$ and $\text{sPar}(I)$.

- (c) There are $\mathcal{P}, \mathcal{Q} \in \text{Par}(I)$ such that $\mathcal{P} \succ_{\text{ref}} \mathcal{Q}$ but $\mathcal{P} \not\succ_{\text{mesh}} \mathcal{Q}$.
- (d) There are $\mathcal{P}, \mathcal{Q} \in \text{Par}(I)$ such that $\mathcal{P} \succ_{\text{mesh}} \mathcal{Q}$ but $\mathcal{P} \not\succ_{\text{ref}} \mathcal{Q}$.
3. Let I be a compact interval and $f \in F(I; \mathbb{R})$. Then the following are equivalent:
- (a) f is Riemann integrable.
- (b) For any $\varepsilon \in \mathbb{R}_{>0}$, there are step functions $\underline{\sigma}, \bar{\sigma} : I \rightarrow \mathbb{R}$ such that
- for any $x \in I$, $\underline{\sigma}(x) \leq f(x) \leq \bar{\sigma}(x)$, and
 - $\int_I \bar{\sigma}(x) dx - \int_I \underline{\sigma}(x) dx < \varepsilon$.
- (c) f is bounded and for any $\varepsilon \in \mathbb{R}_{>0}$, there is a $\mathcal{P} \in \text{Par}(I)$ such that $D(f; \mathcal{P}) < \varepsilon$.
- (d) f is bounded and there is a sequence $\mathcal{P}_\bullet : \mathbb{Z}_{\geq 1} \rightarrow \text{Par}(I)$ such that $\lim_{n \rightarrow \infty} D(f; \mathcal{P}_n) = 0$.
4. Let I be a compact interval and $f \in F(I; \mathbb{R})$. Then
- (a) f is Riemann integrable iff for any $\alpha, \beta \in I$ with $\alpha < \beta$, the restriction $f|_{[\alpha, \beta]}$ is Riemann integrable.
- (b) f is Riemann integrable iff for any $\phi \in C^0(\mathbb{R}; \mathbb{R})$, $\phi \circ f$ is Riemann integrable.
5. Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. Consider the following statements:
- I: $f_\bullet \rightarrow f_\infty$ uniformly.
- II: $f_\bullet \rightarrow f_\infty$ pointwise.
- III: f_∞ is Riemann integrable.

IV: f_∞ is Riemann integrable and

$$\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx.$$

V: For each $n \in \mathbb{Z}_{\geq 0}$, $|f_n - f_\infty|$ is Riemann integrable and

$$\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0.$$

Then

- (a) If **I**, then **IV** and **V**.
- (b) If **II**, then **III**.
- (c) If **II** and **III**, then **IV**.
- (d) If **II** and **IV**, then **V**.

6. Let I be a compact interval. Then

- (a) If $f \in F(I; \mathbb{R})$ is nondecreasing or nonincreasing, then f is of bounded variation.
- (b) $BV(I; \mathbb{R})$ is the smallest subset of $F(I; \mathbb{R})$ that contains all functions of the form

$$\sum_{i=1}^n c_i f_i,$$

where $n \in \mathbb{Z}_{\geq 1}$, $c_1, c_2, \dots, c_n \in \mathbb{R}$ and $f_1, f_2, \dots, f_n \in F(I; \mathbb{R})$ are all nondecreasing.

- (c) If $f \in F(I; \mathbb{R})$ is of bounded variation, then it is Bourbaki integrable.
- (d) If $f \in F(I; \mathbb{R})$ satisfies the following property, then it is both continuous and of bounded variation:

$$\exists \alpha \in \mathbb{R}_{>0}, \exists C \in \mathbb{R}_{>0}, \forall x_1, x_2 \in I : \\ |f(x_1) - f(x_2)| \leq C|x_1 - x_2|^\alpha.$$

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/cdZsheM5HZSaU41S6>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1212058/assignments/7390767>,

see the Gradescope [documentation](#) for instructions.

5 When to Submit

This problem set is due on February 27, 2026 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.